

e-ISSN: 3048-4758

Volume 09 Issue 02 May-Aug, 2026

***Corresponding Author:
Vedant Pathrabe,**

*Student, Department of
Electronics &
Telecommunication
Engineering, Priyadarshini
Bhagwati College of
Engineering, Nagpur,
Maharashtra, India*

Submission Date: Mar 24,
2026

Copyright Received Date:
Mar 27, 2026

Published Date: Apr 27,
2026

Cite as:

*Reshma Satpute, Vedant
Pathrabe, Raksha
Sonewane, Lavanya
Gabhane, Mahak Gupta, &
Dr. J.Y. Hande. (2026).
Design and Implementation
of a Voice-Controlled AI
Robot (VOCIBOT). Journal
of Advancement in
Communication System,
9(2), 34–45.
<https://doi.org/10.5281/zenodo.19817098>*

Design and Implementation of a Voice-Controlled AI Robot (VOCIBOT)

**Reshma Satpute¹, Vedant Pathrabe^{2*}, Raksha Sonewane³,
Lavanya Gabhane⁴, Mahak Gupta⁵, Dr. J.Y. Hande⁶**

¹⁻⁵Student, Department of Electronics &
Telecommunication Engineering, Priyadarshini Bhagwati
College of Engineering, Nagpur, Maharashtra, India

⁶Assistant Professor, Department of Electronics &
Telecommunication Engineering, Priyadarshini Bhagwati
College of Engineering, Nagpur, Maharashtra, India

ABSTRACT

This paper presents the design, implementation, and experimental validation of a voice-controlled AI robotic system named VOCIBOT. The proposed system enables hands-free robotic navigation using AI-based speech recognition combined with embedded control and wireless communication. Voice commands are processed externally using a mobile application and transmitted to the robotic unit via Bluetooth. An Arduino-based controller executes validated commands to drive motors, while an ESP32-CAM module provides real-time visual feedback. Experimental results demonstrate low response latency, reliable command execution, and stable indoor performance. The proposed system offers a cost-effective and scalable solution for assistive, surveillance, and educational robotic applications.

Keywords:- Voice-Controlled Robot, VOCIBOT, Embedded Systems, Arduino Uno, ESP32-CAM, Bluetooth, Human Robot Interaction

INTRODUCTION

Advancements in artificial intelligence and embedded electronics have significantly transformed the field of human–robot interaction. Prior research demonstrates that traditional robotic systems often depend on manual controllers, limiting accessibility and flexibility [1], [2]. Voice-based control has emerged as an intuitive alternative, enabling hands-free interaction and improved usability, particularly for assistive and service robotics applications [3], [4].

Several studies report that the integration of speech recognition with embedded controllers improves human–robot interaction by reducing user effort and response time [4], [6]. However, most existing implementations focus on either basic motion control or conceptual system design, with limited emphasis on real-time monitoring and safety mechanisms [7], [8].

Motivated by these limitations, this research focuses on the practical realization of a voice-controlled robotic system emphasizing real-time responsiveness, operational safety, and visual feedback. Unlike survey-based studies such as [6]–[10], this work concentrates on system design, hardware integration, control logic implementation, and experimental validation, using low-cost embedded components.

SYSTEM OVERVIEW

The proposed VOCIBOT system is developed using a modular and layered architecture, which is widely adopted in embedded robotic systems to improve design flexibility, scalability, and ease of maintenance [4], [6]. By separating the system into independent functional modules, hardware and software upgrades can be implemented without affecting the overall system operation. This approach is particularly beneficial for prototyping and research-oriented robotic platforms.

A key design feature of the system is the use of external AI-based speech recognition instead of on-board voice processing. Speech recognition is performed on a mobile device using an AI-enabled application, and only the processed control commands are transmitted to the robotic unit.

This strategy significantly reduces the computational and memory load on the embedded controller, enabling faster response times and improved system stability when compared to fully on-device speech recognition systems [7], [9].

Wireless communication between the user interface and the robot is established using Bluetooth technology, which has been extensively used in indoor robotic control applications due to its low latency, low power consumption, and implementation simplicity [6], [7]. Bluetooth communication ensures reliable command transmission within a short range, making it suitable for indoor navigation and experimental environments where real-time interaction is essential.

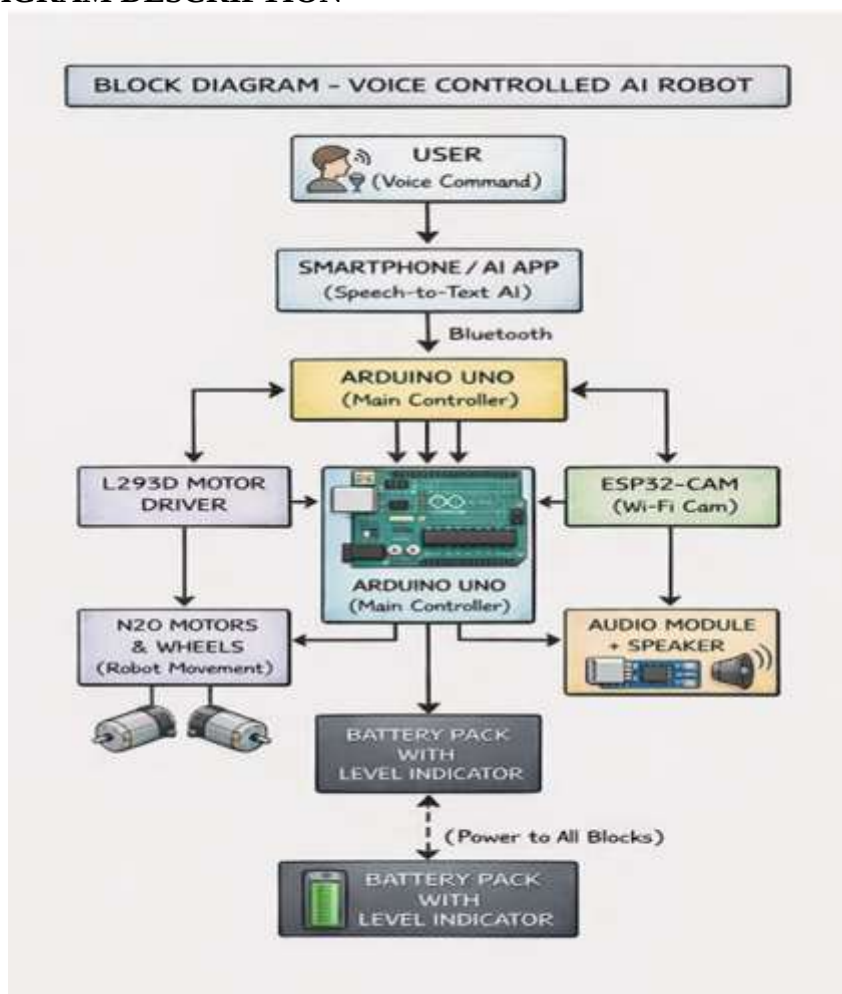
To enhance operational reliability, the system integrates real-time visual monitoring and safety mechanisms. An ESP32-CAM module provides live video streaming, allowing users to remotely observe the robot's surroundings during operation.

This feature improves situational awareness and enables safer remote control, especially in environments with limited visibility [10]. Additionally, an ultrasonic sensor is employed for continuous obstacle detection. Distance measurements are analyzed before executing motion commands, preventing collisions and improving navigation safety, as recommended in prior robotic safety studies [8].

Overall, the VOCIBOT system extends beyond conventional voice-controlled robots by combining external intelligence, embedded control, wireless communication, visual feedback, and safety features into a unified platform.

This integrated architecture enhances usability, reliability, and real-time performance, making the system suitable for assistive, educational, and experimental robotic applications.

BLOCK DIAGRAM DESCRIPTION



The block diagram of the proposed VOCIBOT system illustrates the functional interaction between hardware and software modules responsible for voice processing, communication, control, actuation, monitoring, and safety. Each module is designed to operate independently while maintaining synchronized coordination to achieve reliable voice-controlled robotic navigation.

Voice Command Processing Unit

The voice command processing unit is responsible for acquiring and interpreting user voice input. In the proposed system, voice commands are captured using a smartphone equipped with an AI-based speech recognition application. The spoken commands are converted into text and subsequently mapped to predefined robotic control instructions such as forward, backward, left, right, and stop.

Offloading speech recognition to an external device significantly improves recognition accuracy and system responsiveness, as mobile platforms provide higher computational capability and advanced AI models compared to low-cost embedded controllers. This approach also eliminates the need for complex signal processing on the microcontroller, thereby reducing system complexity and power consumption. Similar external speech processing architectures have been successfully employed in voice-controlled robotic systems reported in previous studies [7], [9].

Wireless Communication Unit

The wireless communication unit enables command transmission between the user interface and the robotic platform. In this system, an HC-05 Bluetooth module is used to establish a serial wireless communication link. Bluetooth technology is selected due to its low implementation cost, ease of configuration, and suitability for short-range indoor robotic applications.

Once the voice command is converted into a control instruction, it is transmitted via Bluetooth to the embedded controller. Bluetooth-based communication ensures minimal latency, enabling near real-time command execution. Prior research has demonstrated that Bluetooth modules provide stable and reliable connectivity for voice-controlled robotic systems operating in indoor environments [6], [7].

Embedded Control Unit

The embedded control unit serves as the core processing module of the VOCIBOT system. An Arduino Uno microcontroller is employed to receive wireless commands, validate control instructions, and coordinate the operation of actuators and sensors. The microcontroller executes a predefined control algorithm that interprets incoming commands and generates appropriate output signals.

Arduino-based platforms are widely used in research and educational robotics due to their simplicity, affordability, and extensive software support. The use of Arduino enables rapid development and easy integration of multiple peripherals, making it suitable for experimental robotic prototypes. Similar embedded control strategies have been reported in voice-controlled and Bluetooth-operated robotic systems [4], [6].

Motor Actuation Unit

The motor actuation unit converts control signals generated by the microcontroller into physical motion. The L293D motor driver IC is used as an interface between the Arduino and the DC motors. Since microcontroller output pins cannot directly drive motors due to current limitations, the motor driver amplifies the control signals and provides bidirectional motor control.

By selectively enabling driver inputs, the system achieves precise control over motor direction and speed, allowing smooth navigation of the robotic platform. The L293D motor driver is commonly used in low-power robotic systems and has been proven effective in similar robotic implementations [4], [6].

Monitoring and Safety Unit

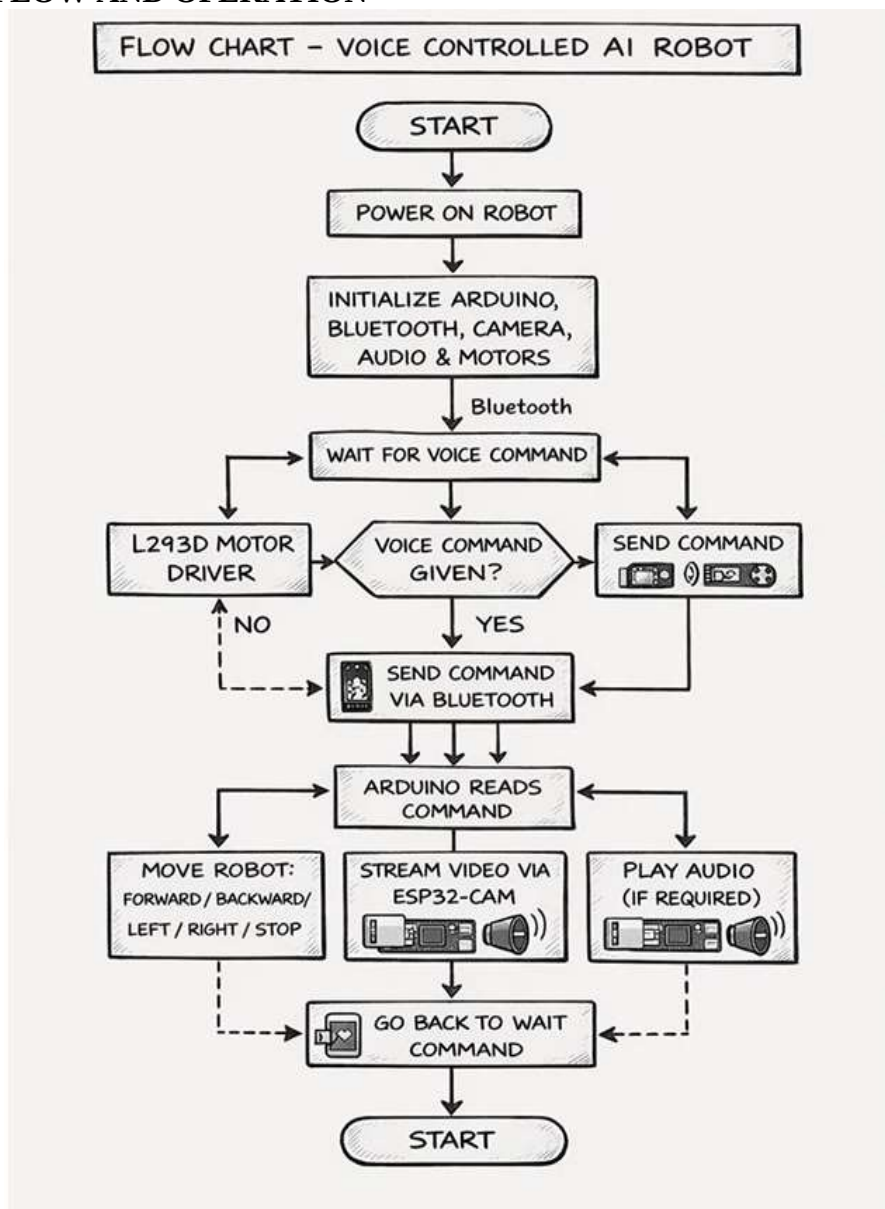
The monitoring and safety unit enhances operational awareness and prevents accidental collisions. An ESP32-CAM module is integrated into the system to provide real-time video streaming. Live video feedback allows the user to remotely monitor the robot's environment and make informed control decisions, particularly in areas with limited visibility or restricted access [10].

In addition to visual monitoring, an ultrasonic sensor is employed for obstacle detection. The sensor continuously measures the distance between the robot and nearby objects. When an obstacle is

detected within a predefined threshold distance, the control unit restricts or overrides motion commands to prevent collisions. Ultrasonic-based obstacle

detection has been widely recognized as a reliable and cost-effective safety mechanism for indoor robotic navigation [7], [8].

SYSTEM FLOW AND OPERATION



The operational workflow of the proposed VOCIBOT system is based on a distributed processing architecture, in which computationally intensive speech recognition tasks are executed externally, while the embedded controller is dedicated to real-time decision-making and actuation [7], [9]. This separation of processing

responsibilities improves system efficiency and ensures consistent real-time performance on resource-constrained embedded hardware.

The operation begins with system initialization, during which all hardware modules including the microcontroller,

Bluetooth interface, motor driver, ultrasonic sensor, and camera module are powered and configured. The Bluetooth module then enters a pairing mode and waits for successful connection with the user's mobile device. Once pairing is established, the system transitions into a command-listening or idle state, indicating readiness to receive voice-based instructions.

When a voice command is issued by the user, speech recognition is performed on the mobile device, and the corresponding control instruction is transmitted wirelessly to the robotic platform. Upon reception, the Arduino microcontroller performs command validation by checking the instruction against a predefined command set. This validation process prevents the execution of unintended or corrupted commands, an approach widely recommended in safety-aware embedded robotic systems [4], [7].

Before executing the received command, the controller simultaneously evaluates sensor inputs, particularly data from the ultrasonic sensor. If an obstacle is detected within a predefined safety threshold, the controller either modifies or blocks the execution of motion commands, thereby reducing the risk of collision. This conditional execution mechanism introduces a safety layer into the control flow, extending functionality beyond basic voice-controlled motion systems [8].

During motion execution, the motor driver actuates the Gear motors according to the validated control signals. At the same

time, the ESP32-CAM module continuously streams real-time video data, providing visual feedback to the user. This parallel operation of motion control and visual monitoring enables informed remote operation and improves situational awareness, particularly in indoor or confined environments [10].

After the completion of each command, the system returns to the command-listening state, allowing continuous and interactive operation. This cyclic operational flow ensures responsive human-robot interaction while maintaining safety and system stability.

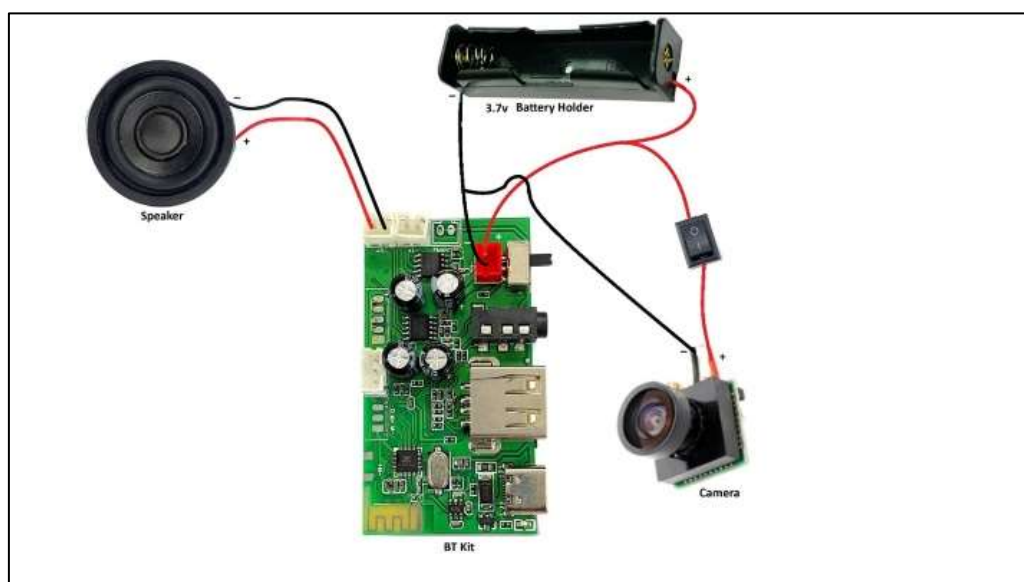
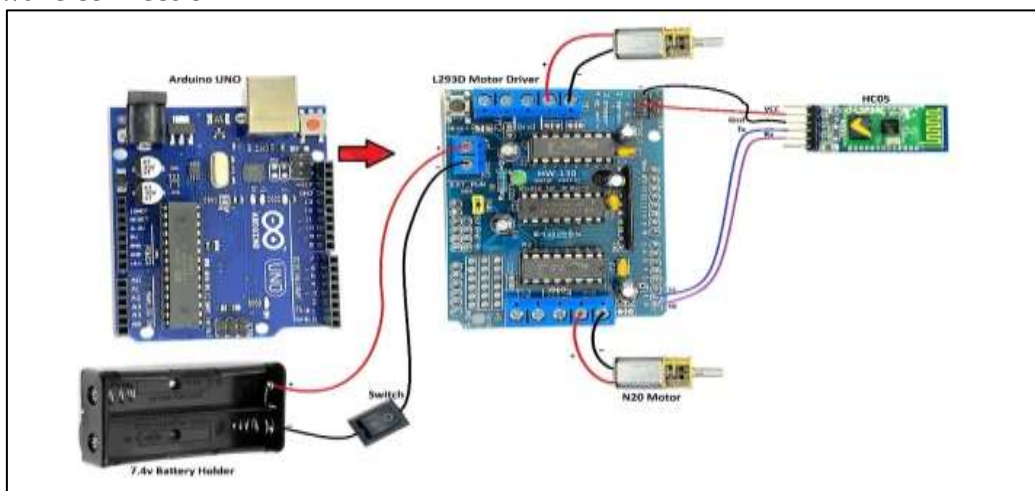
The overall flow architecture is consistent with previously reported Bluetooth-based voice-controlled robotic systems [6]–[9] but extends them through the integration of real-time visual feedback and obstacle-based safety intervention, thereby enhancing reliability and practical usability.

HARDWARE AND SOFTWARE IMPLEMENTATION

Hardware Components

The hardware architecture of the proposed VOCIBOT (Voice-Controlled Intelligent Bot) is designed to provide reliable performance, modular integration, and efficient power management. The system integrates multiple embedded components including a microcontroller, wireless communication modules, actuators, power management units, and audio-visual devices to enable intelligent robotic interaction.

Hardware connection



Software Implementation

The software architecture of the VOCIBOT system consists of embedded firmware running on the microcontroller and external software applications responsible for voice processing and artificial intelligence integration.

Embedded Firmware (Arduino IDE)

The embedded firmware is developed using the Arduino Integrated Development Environment (IDE) and programmed in Embedded C. The firmware controls the robot's movement by interpreting commands received through the Bluetooth communication module.

The firmware performs several tasks including:

- Serial communication with Bluetooth module
- Command decoding
- Motor driver control
- Movement execution

The robot follows command-based navigation logic where each movement corresponds to a predefined instruction.

Example control commands include:

- Forward movement
- Backward movement

- Left turn
- Right turn
- Stop

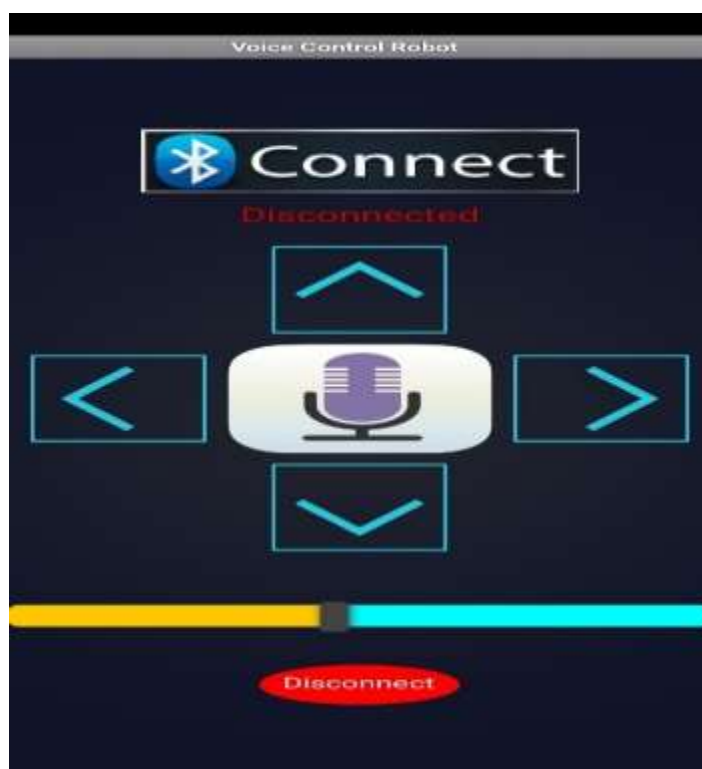
This simple yet effective command-based architecture ensures fast response and reliable control.

Mobile Voice Control Application

A mobile voice control application is used to convert spoken commands into digital

control instructions. The application utilizes the smartphone's built-in speech recognition capabilities to process user voice input.

Once the voice command is recognized, the application maps it to a predefined control signal and sends it to the robot through the Bluetooth module.



The mobile application interface allows users to:

- Connect to the robot via Bluetooth
- Issue voice commands
- Control robot movement
- Adjust movement speed

The user interface of the application is shown in the figure above, where directional controls and a microphone button enable voice-based navigation.

AI Interaction using Gemini

For intelligent conversational interaction, the system integrates Google Gemini AI

through an external device such as a smartphone or computer. Gemini processes user queries and generates appropriate responses using natural language processing.

When the user asks a question, the mobile application sends the request to the Gemini AI system. The generated response is converted into speech and played through the robot's speaker system.

This integration enables the robot to function not only as a remotely controlled vehicle but also as an AI-powered

interactive assistant capable of answering questions and providing information.

voice-controlled navigation and operational safety.

EXPERIMENTAL RESULTS AND PERFORMANCE ANALYSIS

The VOCIBOT system was experimentally evaluated for real-time

Key performance metrics include command response time, communication reliability, navigation accuracy, and obstacle detection.



Command Response Time
Command response time refers to the delay between issuing a voice command and motor activation.

Experimental results show an average response time of **< 500 ms**, achieved through external speech processing and efficient embedded control.

Wireless Communication Reliability

The HC-05 Bluetooth module maintained stable communication within an indoor range of **8–10 meters**, with negligible delay or packet loss, confirming its suitability for short-range applications.

Navigation

The system achieved an average navigation accuracy of **~95%** in low-noise

Accuracy

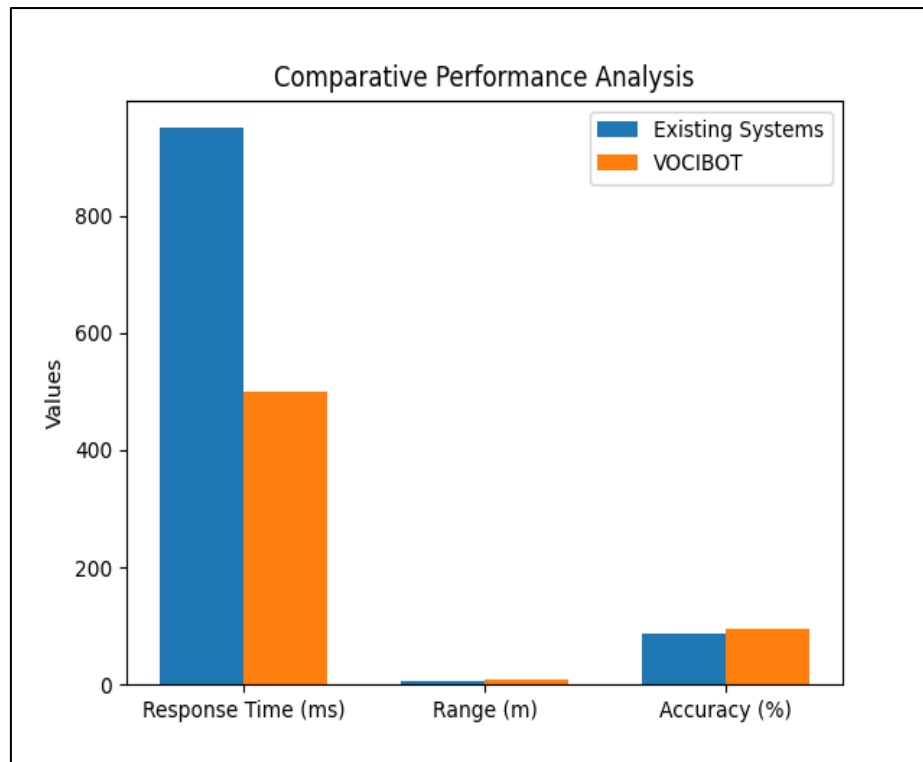
environments. This performance is supported by reliable speech recognition and effective command execution.

Comparative

Compared to earlier systems, VOCIBOT demonstrates improved response time, higher accuracy, and enhanced safety due to integrated obstacle detection and real-time monitoring.

Analysis

Parameter	Existing Systems [6], [7]	Proposed VOCIBOT
Response Time	700–1200 ms	< 500 ms
Communication Range	5–8 m	8–10 m
Navigation Accuracy	85–90%	95%
Real-Time Monitoring	Not Available	Available



Overall, VOCIBOT offers a cost-effective, modular, and efficient solution suitable for assistive, educational, and surveillance applications, contributing to the advancement of intuitive human–robot interaction systems.

REFERENCES

1. International Conference on Advances Automotive Technologies, 2016.
2. T. Lens, J. Kunz, and O. von Stryk, "Bio Rob-Arm," International Symposium on Robotics, 2010.
3. "A Smart Voice Controlled Pick and Place Robot," IRJET. "Arduino Based Voice Controlled Robot," IRJET.
4. "Wireless Control Robotic Arm Using an Android Application," IJAREEIE. [6] "Voice Controlled Robot using Bluetooth," IJERT.
5. "Speech Recognition Wireless Control System for Mobile Robotics," MDPI.
6. "Voice Controlled Application using ESP32," Journal of Electrical Engineering and Automation.
7. "Acoustic Robot Using Arduino," *International Journal of Engineering Research & Technology (IJERT)*, 2017.
8. "Arduino Based Voice Controlled Robot Vehicle," *IOP Conference Series: Materials Science and Engineering*.
9. "Arduino Based Bluetooth Voice Controlled Robot Car and Obstacle Detector," *International Journal of Scientific Research in Science, Engineering and Technology*.
10. "Voice Controlled Robot Using Bluetooth," *International Journal of Engineering Research & Technology* "A Review on Voice Controlled AI Robot (VOCIBOT)" *International Journal of Engineering Research and Development, Volume e 22, Issue 2 (February 2026)*.